

Directional Partial-Wave Sweeps and the Krylov Solve

The propagator matvec for elastic multiple scattering on a depth-stratified voxel grid

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Abstract

The multiple scattering of elastic waves by voxels superimposed on a depth-stratified background is a Foldy–Lax / volume integral equation, $T = T_0(I - G_0 T_0)^{-1}$, solved by a Krylov method whose only nontrivial operation is the application of the background propagator G_0 . This note describes that application as an alternating-direction partial-wave summation: the lateral coupling is summed by constant- z left-right (k_x) and constant- (z, x) in-out (k_y) cumulative-phase sweeps, and the coupling between depth planes by the up-down Kennett reflectivity recursion (k_z). Choosing the lateral directions for the in-plane coupling sums the propagator along axes in which the voxels are separated, so the k_z partial-wave sum that diverges at zero depth separation is never evaluated. The sweeps are a pure forward matvec; all multiple scattering, to every order, is resolved by the Krylov iterations. A sparsified local system with centre-anchored stencils preconditions those iterations and needs no knowledge of the layering, but at physical contrasts, on the lattices measured, the solve converges in a few iterations without it and the gain is marginal; sweeping that system in place of solving it requires an approximation of the Dirichlet-to-Neumann map carried by its Schur complements.

Companion series. *One of a companion series on elastic multiple scattering from cubic heterogeneities on a depth-stratified background, the T -matrix programme $T = T_0(I - G_0 T_0)^{-1}$: (I) the stratified inter-site propagator, the same-depth convergence theory, the SH and P/SV traction kernels, interplane side scattering and the first-shell near stencil (`EwaldIntraPlanePropagator`); (II) the contrast-source volume integral equation and the CG-FFT / seismic grid T -matrix lineage (`ContrastSourceVIE`); (III) the directional partial-wave sweep matvec and the Krylov multiple-scattering solve (`DirectionalSweepSolver`); with the empirical companion `RQ1_Planar-Decomposition-Proposal`. This is note (III).*

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1 The system, and where the cost lives

Voxels (cubic cells) are superimposed on a background that carries the lateral average of the medium at each depth: laterally homogeneous within a constant- z plane, layered in z . Each voxel has an on-site T-matrix T_0 , the single-cell scatterer (Eshelby with the $r = 0$ delta-function correction up to the 27-component Galerkin form, the self-term already closed), and the voxels are coupled by the background propagator G_0 . The multiple scattering is the Foldy–Lax system [1, 2, 3]

$$(I - G_0 T_0) \psi = \psi^{\text{inc}}, \quad T = T_0 (I - G_0 T_0)^{-1}, \quad (1)$$

where ψ collects the exciting (incoming) fields at the voxels. The operator T_0 is block-diagonal and local. The whole cost is the off-diagonal G_0 : the inter-voxel communication. The strategy is to solve (1) by a Krylov method whose matrix-vector product is one application of G_0 , and to make that application fast.

The essential point is the division of labour. The application of G_0 is a *forward* operation, a summation of partial waves that carries the radiation of every voxel to every other. It contains no inversion and no embedding. All of the multiple scattering, the reverberations to every order, is built and resummed by the Krylov iterations, not by the propagator.

2 The matvec is a partial-wave summation

Given source amplitudes $b = T_0 \psi$ at the voxels, the matvec returns the field they radiate, $\psi' = G_0 b$. For the homogeneous part of the background the propagator is a sum of partial waves, and a partial wave’s phase accumulates linearly in space: e^{ikx} at the next voxel is the current one times e^{ikp} , with p the pitch. The sum is therefore built by carrying that running phase along a line, one term added per voxel. This is the sweep. The only freedom is which axis to sum along, and the choice is dictated by where the voxels are separated.

2.1 Lateral sweeps: constant- z left-right and constant- (z, x) in-out

Within a depth plane the background is homogeneous, so the lateral propagation is pure phase, with no background reflections to sum. Split the in-plane propagator on the k_x pole into a right-going and a left-going partial wave and sum each by a sweep in its own direction. With a^+ the right-going accumulated field, s_n the source radiated at voxel n , and $\Phi_x = e^{ik_x p}$ the one-pitch lateral propagator,

$$a_{n+1}^+ = (a_n^+ + s_n) \Phi_x \quad (\text{left to right}), \quad a_{n-1}^- = (a_n^- + s_n) \Phi_x \quad (\text{right to left}). \quad (2)$$

Each partial wave is carried along the direction in which it decays: the right-going evanescent branch $e^{-\kappa_x x}$ is swept to the right, the left-going branch to the left, so the growing exponential is never formed and no reflection/transmission bookkeeping is needed for stability. The in-out sweep in y is identical, run along each constant- (z, x) line on the k_y pole. The two together sum the full in-plane partial-wave series as nested one-dimensional phase accumulations.

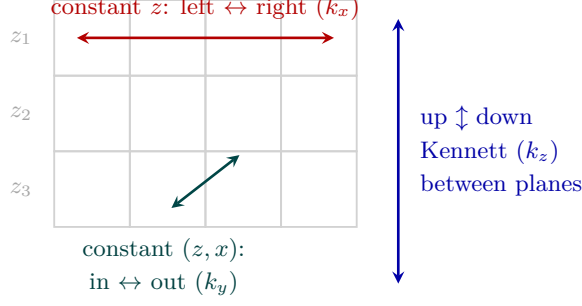


Figure 1: The G_0 matvec as three directional partial-wave sweeps over the voxel grid (cross-section; the third axis y runs into the page). Within each depth plane the lateral coupling is summed by constant- z left-right (k_x) and constant- (z, x) in-out (k_y) cumulative-phase sweeps; the coupling between depth planes is the up-down Kennett reflectivity recursion (k_z). The lateral directions carry the in-plane coupling along axes of nonzero separation, so the k_z sum is never evaluated at $\Delta z = 0$.

2.2 Vertical sweep: up-down Kennett between depth planes

In z the background is not homogeneous: the lateral average changes from plane to plane, so the depth-average layering has its own reflections and transmissions. The coupling between depth planes is therefore the up-down Kennett reflectivity recursion [4] on the k_z pole, propagating the source amplitudes through the layered background and summing its internal up/down multiples. This is the one sweep that carries reflection and transmission coefficients, and it does so for the *background* layering, not for the heterogeneity; it is the standard invariant-embedding recursion, stable by construction, and its coefficients depend only on the background and are formed once. The heterogeneity scattering remains entirely in T_0 and the Krylov loop.

2.3 Why these directions

The conventional construction sums the in-plane coupling on the k_z pole, the up/down decomposition, whose surviving (k_x, k_y) integral loses its convergence factor $e^{-\kappa_\perp |\Delta z|}$ at $\Delta z = 0$ and diverges at equal depth. The lateral split avoids it outright: the in-plane coupling is summed on the k_x and k_y poles, along which distinct in-plane voxels are separated, so the divergent same-depth k_z sum is never the one evaluated. The k_z sweep is used only between planes, where $\Delta z \neq 0$ and the Kennett recursion is exact and stable. Every sweep marches along an axis of nonzero separation.

2.4 The same-depth term in a stratified background

The lateral sweep of §2.1 sums the same-depth coupling of the *whole space*. That is not the whole of it. When the background is stratified, two voxels at the same depth are also coupled by paths that leave one voxel, reflect from a layer boundary above or below, and return to the plane. Those paths are absent from the lateral sweep by construction, because the k_x residue carries only the direct term.

They cannot simply be added by evaluating the stratified propagator at $\Delta z = 0$ and integrating: that is the divergence of §2.3, and it is present in the stratified kernel exactly as in the whole-space one, its strain-strain block growing like $|k_x|$. The resolution is to subtract it. Writing G_0^{lay} for the stratified propagator between a plane and itself and G_0^∞ for the whole-space kernel at equal depth,

$$G_0^{\text{lay}}(\Delta z = 0) = \underbrace{G_0^\infty(\Delta z = 0)}_{\text{lateral sweep, by } k_x \text{ residue}} + \underbrace{\Delta G_0}_{\text{reverberation only}}, \quad (3)$$

and the two terms on the right are individually non-integrable while their difference is not. Every path contributing to ΔG_0 travels at least twice the distance H to the nearest interface, so it carries $e^{ik_z 2H}$ and ΔG_0 decays exponentially in k_x . Its k_x integral therefore converges, and the divergence cancels between the two terms rather than being regularised in either.

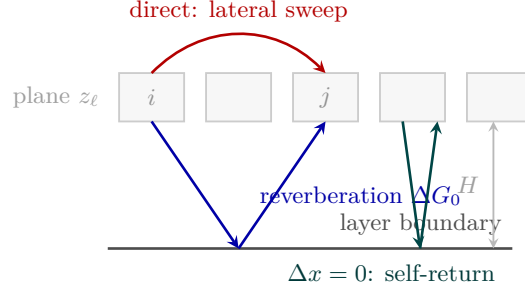


Figure 2: The same-depth interactions of (3), at one depth plane. The **direct** whole-space coupling between voxels i and j is summed by the lateral sweep, on the k_x pole, exactly. The **reverberation** leaves the plane, reflects from a layer boundary a distance H away and returns to it; the lateral sweep cannot carry it, and it is what ΔG_0 supplies. Both terms diverge on their own — each grows like $|k_x|$ — and their difference does not, because every reverberating path travels at least $2H$ and so carries $e^{ik_z 2H}$. The **third path** is the same process returning to the *source voxel*: it is a genuine self-interaction, it is not contained in T_0 (which is the single-site T-matrix of the whole space), and dropping it — the natural move, since the whole-space self-term is excluded everywhere else — would silently discard every layer-return-to-self.

Two consequences are worth stating plainly. First, ΔG_0 vanishes identically when the background is uniform, so the construction reduces to the whole-space sweep with nothing left over. Second, ΔG_0 is retained at $\Delta x = 0$: a voxel *does* interact with itself through the layering, and since T_0 is the single-site T -matrix of the whole space, its self-term does not contain that path. Dropping the $\Delta x = 0$ entry — the natural thing to do, since the whole-space self-term is excluded everywhere else — would silently discard every layer-return-to-self.

3 Alternating-direction composition

The three sweep pairs compose by alternating direction, in the spirit of the alternating-direction methods of Peaceman and Rachford [5]: the two-dimensional in-plane sum factors into an x pass and a y pass, and the full three-dimensional propagator into those plus the z recursion. Each pass is a one-dimensional, $O(N)$ accumulation, and the matvec is their ordered application:

1. **up-down** (k_z , **Kennett**): propagate the source amplitudes between depth planes through the layered background;
2. **left-right** (k_x): along each constant- z row, sweep left to right for the right-going field and right to left for the left-going field;
3. **in-out** (k_y): along each constant- (z, x) line, sweep in and out the same way.

The result is $\psi' = G_0 b$, the field every voxel radiates onto every other, summed along directions of nonzero separation throughout. Six direction-pure sweeps in all, none of which ever forms the $\Delta z = 0$ sum.

4 The Krylov solve

The matvec of Section 3 is a pure forward application of G_0 . The multiple scattering is solved around it by a Krylov method (GMRES [6]) applied to (1). One iteration is

$$v \mapsto v - G_0 T_0 v, \quad (4)$$

that is, one block-diagonal T_0 (local, the cube T-matrix) followed by one G_0 (the three sweeps). The Krylov subspace builds the scattering series and resums it, reaching the solution in far fewer applications than the bare Neumann (Born) series and remaining stable at strong contrast where that series would diverge. The reverberations and back-and-forth that the forward sweep deliberately does *not* contain are produced here, across the iterations.

For strong contrast the iteration is preconditioned, the role played by the renormalised scattering series of Jakobsen and Wu [7] and the convergent Born series of Osnabrugge, Leedumrongwatthanakun and Vellekoop [8]: a preconditioner on the Krylov solve, transparent to the sweep, which never needs to know the contrast level.

The stack, end to end.

1. **Setup.** Form T_0 per voxel (self-term closed) and the background Kennett reflectivity coefficients (once).
2. **Matvec** $G_0 b$. Up-down Kennett between planes, then left-right (k_x) and in-out (k_y) cumulative-phase sweeps within planes.
3. **Solve.** GMRES on $(I - G_0 T_0)\psi = \psi^{\text{inc}}$, preconditioned at strong contrast, each step one T_0 and one matvec.
4. **Output.** The exciting fields ψ , hence the global T and any observable (reflection matrix, shot gather) by one more radiation of $b = T_0 \psi$.

5 Validation status

The 2½-D case (heterogeneity in (z, x) , y invariant, k_y a parameter) is implemented and gated. Each claim below is the same physics through two independent code paths, so the residual measures defect, not discretisation, except where noted.

Claim	Arbiter	Residual	Evidence
amp $\times$ phase ⁿ = bundled kernel	k_x -residue kernel	1.5×10^{-15}	<code>t:sweep_kernels</code>
residual <i>flat</i> in separation $n = 1 \dots 32$	—	—	<code>t:sweep_kernels</code>
left amplitude = x -reflection of right	parity identity	1×10^{-15}	<code>t:sweep_kernels</code>
lateral sweep = pairwise sum, distinct sources	$O(N^2)$ sum	1.5×10^{-13}	<code>g:sweep_rung2_lateral</code> [2]
residual <i>flat</i> in lattice size	—	—	<code>t:directional_sweeps</code>
k_y -integrated kernel = closed form	Kupradze	falls to 2×10^{-5}	<code>g:sweep_rung2_lateral</code> [2b]
inter-plane sweep = pairwise sum	Kupradze	falls to 8×10^{-4}	<code>g:sweep_rung3_vertical</code> [3]
lateral coupling does not wrap	kernel at true sep.	0	<code>g:sweep_rung3_vertical</code> [3b]
stratified operator reduces to whole space	k_z -residue kernel	1.1×10^{-15}	<code>g:sweep_rung3_layered</code> [3L-a]
reverberation vanishes, uniform background	—	$< 10^{-15}$	<code>t:directional_sweeps</code>
self-energy $\equiv$ dressed	direct vs dressed	4×10^{-16}	<code>g:sweep_dressed_equivalence</code> [D1]
$T_d = T_0(I - DT_0)^{-1}$	J_6 law, $\pm k$	8×10^{-14}	<code>g:sweep_dressed_equivalence</code> [D3]
reverberation obeys symplectic reciprocity	—	$10^{-4} \rightarrow 10^{-12}$	<code>g:sweep_dressed_equivalence</code> [D2]
self-energy recedes with the reflector	support count	exact	<code>t:directional_sweeps</code>
partition: every pair once, self-term	two direct	1.2×10^{-12}	<code>g:sweep_rung5_gmres</code>
GMRES = dense LU = Neumann, $\rho(G_0 T_0) = 0.92$	methods		
ΔG_0 lies in the rank-3 one-way span	pseudo-inverse	4×10^{-9}	<code>g:dg0_absolute_magnitude</code> [M1]
	residual		
ΔG_0 magnitude, full 9×9 forward	Kennett	2×10^{-10}	<code>g:dg0_absolute_magnitude</code> [M2]
	reflectivity		
weak, strong and multi-interface, $k_{\parallel} h \leq 3$	—	—	<code>g:dg0_absolute_magnitude</code> [M2]
P/S flux factor, predicted not fitted	$i\beta\sqrt{\eta_S}/\alpha\sqrt{\eta_P}$	8 s.f.	<code>g:dg0_absolute_magnitude</code> [M3]
y -integral of 3-D form = $2\frac{1}{2}$ -D	k_z residue vs	4×10^{-13}	<code>g:y_integral_2p5d</code>
propagator	Kupradze		
layered table reduces, uniform background	closed form	1.3×10^{-14}	<code>t:layered_stack_table</code>
residual <i>flat</i> in the transverse rule	—	—	<code>t:layered_stack_table</code>

Evidence column, abbreviated to fit: `g:name` is the standalone script `scripts/gate_name.py`, run directly, and `t:name` is the module `cubic_scattering/tests/test_name.py`, run under `pytest`. A bracketed tag is the check's label within that file, printed verbatim in its output.

Three points of method are worth recording because each cost time to learn. Symmetry and reciprocity invariants are homogeneous of degree one and therefore blind to an overall scale error; every such gate here is paired with one absolute comparison. Where a tolerance had to be loosened it was loosened only after measuring why, and the accompanying test asserts the property that actually discriminates — that the residual is flat in separation, or flat in lattice size — since the dangerous failure is a factor that accumulates along the sweep. Finally, a residual that does *not* fall under refinement is a convention error, not a quadrature error: two of the discrepancies met here were exactly that, one an index ordering and one an attenuation mismatch between an operator and the Green's function it multiplied.

On the self-energy, and which invariant applies to it. The same-depth reverberation ΔG_0 does *not* satisfy the whole-space symmetry $WM = (WM)^T$: measured 0.33 to 1.06 across k_x , at every separation. That is not a defect but a misapplied invariant. At coincident planes the *full* layered kernel fails the symplectic reciprocity law at 9.5×10^{-3} while ΔG_0 satisfies it at 8×10^{-14} : the non-reciprocal part is the one-sided jump carried by the direct term, and removing that term is precisely what (3) does. The reverberation is the clean reciprocal object; the whole-space symmetry is the wrong thing to demand of it. This is recorded because the same pattern — a plausible invariant that the exact reference also fails — has already cost this programme one false gate.

The absolute magnitude of ΔG_0 , now closed. Reciprocity, the vanishing without layering, the recession with the reflector and the dressed- T identity are all homogeneous of degree one and therefore say nothing about scale. The magnitude is settled instead by a *forward* comparison carrying no fitted quantity. At a plane over a layered half-space the reverberation must factor

through the one-way mode maps of §2.4,

$$\Delta G_0 = D_{\uparrow} R_D S_{\downarrow}, \quad (5)$$

with R_D the total downward reflection matrix referenced at the plane — built by the Kennett reflectivity recursion alone, every internal multiple included. Predicting the full 9×9 from (5) and differencing it against the measured ΔG_0 leaves a relative residual of 2×10^{-10} , at weak contrast, at strong contrast, and on a fast/slow/fast stack, out to $k_{\parallel}h = 3$ in the evanescent regime.

Two things had to be measured rather than assumed for that comparison to mean anything. First, the part of ΔG_0 lying *outside* the rank-three one-way span of (5) is 4×10^{-9} or smaller, so the pseudo-inverse extractions used elsewhere in this programme are reading the operator and not a projection of it. Second, the mode-*conversion* channels $P \leftrightarrow SV$ are invisible to any per-mode ratio test, since a similarity $R \mapsto C^{-1}RC$ by a diagonal C leaves every diagonal entry exactly fixed. Compared directly they disagree — and the disagreement is precisely the Kennett P/S flux normalisation, $c = i\beta\sqrt{\eta_S}/\alpha\sqrt{\eta_P}$, which the mode bridge does not carry and the reflectivity recursion does. Written down *a priori* it reproduces the measured factor to eight significant figures, identically across all three models, and collapses the forward residual from 0.76 to the in-span floor. That is a convention reconciled, not a constant tuned.

The difference is not cosmetic. The mode bridge normalises polarisations to unit *displacement*, and bilinearly, so that they continue analytically into the evanescent regime. The reflectivity recursion normalises to unit *energy flux*, because that is what makes the interface matrices symmetric and unitary — $T_u = T_d^T$ — and without those identities the recursion is not valid. Vertical flux goes as $\rho v^2 \eta |A|^2$, so unit-flux amplitude goes as $1/(v\sqrt{\rho\eta})$, which is where the velocity and the $\sqrt{\eta}$ come from; the factor i on SV is the eigenvector convention of the recursion itself. This same similarity already appears in this package, converting the slab solver’s reflection matrix to the Kennett convention — the sweep solver needed a reconciliation the slab solver had already made, and the two agreeing is corroboration rather than coincidence.

6 The three-dimensional case, and why it is not six sweeps

The architecture above is a 2½-D result. Extending it to three dimensions by giving each sweep a transverse-wavenumber quadrature — the obvious reading, and the one this programme’s design note originally took — does not work, and the reason is worth stating because it is quantitative rather than aesthetic.

A sweep carries a running accumulator indexed by its transverse quadrature nodes. In 2½-D that accumulator is $(n_z, n_{k_z}, 9)$, a few megabytes, because y is invariant and k_y is a single parameter. In three dimensions y becomes a lattice axis *and* the transverse rule becomes two-dimensional, so the accumulator is $(n_z, n_x, n_k, 9)$. At the converged transverse rule — $k_r p = 30$, $\Delta k = 0.156$, reaching 3.2×10^{-7} against the closed form — that is 68 GB on a 16^3 lattice with two live at once, and the inter-plane stack stored spectrally is 615 GB. Both were measured, not estimated.

Nor can a compressed kernel rescue the sweep. A running sweep does not need a quadrature at all, only that the lattice kernel be a short sum of exponentials in the swept offset: a term z_j^m , with m the offset in pitches and z_j a complex ratio, is carried by one amplitude multiplied by z_j per voxel. For an x -then- y sweep the equal-depth propagator $P(mp, np, 0)$ must take the separable form $\sum_{j,l} C_{jl} z_j^m w_l^n$, with the x -ratios z_j shared by every y -offset n and every component, and the accumulator then carries JL amplitudes, J and L the numbers of x - and y -terms. A multi-channel matrix pencil fit of the exact nine-component propagator to 10^{-6} of its largest entry — validated on a synthetic five-term sequence, which it recovers as exactly five terms to 5×10^{-15} , and on random data, which it does not compress — gives no such form. At two points per S wavelength the x -terms never reach 10^{-6} ; at six, they need 15 of the 31

available offsets on a 32-wide lattice and 21 of 63 on a 64-wide one, and the y -stage then never reaches it (`measure_sweep_exponential_rank`). The term count grows with the lattice, so the sweep would be a dense sum in disguise. This is not a weakness of the fit: Engquist and Zhao [10] bound the number of separable terms that represent the Helmholtz Green’s function between two sets to fixed accuracy from below by $k^{s-\delta}$ for any $\delta > 0$ in the geometries they treat, k the wavenumber and s the dimension of the smaller set, with a matching upper bound $k^{s+\delta}$. So no fixed-size exponential representation of the in-plane kernel exists once the lattice spans many wavelengths.

The same operators *tabulated in real space* are 1.2 MB and 149 MB. So in three dimensions the coupling is split by depth separation rather than by direction:

$$G_0 = \underbrace{P(\Delta x, \Delta y, 0)}_{\text{closed form, exact}} + \underbrace{P(\Delta x, \Delta y, \Delta z)}_{\text{closed form}} + \underbrace{\Delta G_0(\Delta x, \Delta y, \Delta z)}_{\text{spectral, transformed once}}, \quad (6)$$

with the whole-space part taken from the closed-form propagator at every separation and *only* the layer reverberation transformed from (k_x, k_y) . That subtraction is the economy of the scheme and not a convenience: the full layered kernel inherits the whole-space kernel’s slow spectral decay and needs some 2.4×10^6 transverse nodes, whereas the reverberation converges at $k_r p = 10$ on about 6.6×10^4 — a factor of 36. The resulting build costs 2.1 h at 16^3 , once, outside the Krylov loop.

The equal-depth closed form in (6) is that of *each plane’s own medium*. The reverberation table subtracts, on its diagonal, the whole-space kernel of the medium at the plane; the closed form added back must be the same one, or the same-plane coupling of a plane in a different material is wrong by the difference of the two whole-space kernels — an error of the order of the material contrast. With every plane in one medium the distinction vanishes, and no configuration measured before this section had planes on both sides of a contrast. The cache builder (`build_g0_cache_3d`) therefore holds one same-depth table per plane, each from that plane’s own medium, and with a layered model the operator does not depend on the caller’s reference medium at all — gated on planes that straddle a contrast, where a single shared table was measured 58% off (`test_apply_g0_3d`).

Correction, 26 September 2026: the $k_r p = 10$ rule holds only far from a contrast.

It was measured on one plane pair in a single medium, with the nearest interface 2.5 pitches away. `scripts/measure_stack_table_cross_material.py` places planes half a pitch from a fast slab, using half-pitch sublayers because a plane may not sit on a material jump. Against the operator’s own scale, $k_r p = 10$ is then off by up to 25%. That happens in cross-material blocks, and *also* in the reverberation (diagonal) block of a plane near the contrast. $k_r p = 20$ is the first rule to converge ($\leq 3 \times 10^{-4}$). A control with no slab is exact at $k_r p = 10$, and halving the node spacing moves nothing, so this is truncation. Subtracting the receiver-plane whole-space kernel leaves more than reverberation once a contrast is near. The cutoff appears to scale with the distance to the nearest contrast rather than with the pitch (consistent with the measurement, not proven). Build cost goes as n_{axis}^2 , so $k_r p = 20$ at fixed spacing costs four times the 2.1 h above. The adjoint gate A1b found the same defect in the layered incident field (note IV).

The (z, x) closed-form y -integral. The two architectures meet exactly, through the y -integral, and it is worth being precise because it is often stated as though 2½-D required a propagator of its own. It does not. For a y -invariant model every voxel is a line of identical voxels along y , so the coupling between two (z, x) columns is the three-dimensional propagator summed along that line,

$$P_{2\frac{1}{2}}(\Delta x, \Delta z; k_y) = \int_{-\infty}^{\infty} P(\Delta x, y, \Delta z) e^{-ik_y y} dy, \quad (7)$$

so the 2½-D propagator is the k_y component of the y -transform of the three-dimensional one, and at $k_y = 0$ it is literally its y -integral. There is no separate 2½-D closed form — only a slice

of the three-dimensional one. Equation (7) is what licenses using the same closed-form table for both geometries, and it is the reason the $\Delta z = 0$ divergence discussed in §2.4 is a property of the k_z -pole construction rather than of the propagator itself.

This is measured, not asserted. Evaluating the left side by the k_z -residue kernel at $k_y = 0$ and inverse-transforming over k_x , against the right side by direct y -quadrature of the Kupradze closed form — two constructions with no derivation in common — agrees to 4×10^{-13} at $\Delta x = 0$ and 1.5×10^{-11} at two pitches (`gate_y_integral_2p5d`). The damping matters: at real ω both integrands decay only algebraically and neither route is clean.

Rebuilding three dimensions: the k_y quadrature. Equation (7) runs one way, from three dimensions to a slice. The step in the other direction is the one the solver actually performs, and it is the inverse transform,

$$P(\Delta x, y, \Delta z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} P_{2\frac{1}{2}}(\Delta x, \Delta z; k_y) e^{ik_y y} dk_y, \quad (8)$$

so a three-dimensional response is assembled by running the 2½-D construction once per k_y node and summing. Every sweep of §2.1 and every Kennett recursion is unchanged inside the integrand; k_y enters only as the parameter it already was. In practice (8) is a trapezoid on a uniform grid symmetric about $k_y = 0$, endpoints half-weighted, with the $1/2\pi$ carried on the quadrature weight.

Unlike the sweeps themselves, this step is quadrature-limited rather than exact, which is why the two rows it governs in §5 are reported as convergence and not as tolerances. Refining the rule from $(k_{y\max}, n_{k_y}) = (60, 1201)$ to $(120, 3201)$ takes the k_y -integrated same-depth kernel from 2.3×10^{-4} to 2.3×10^{-5} against the Kupradze closed form (`gate_sweep_rung2_lateral`), and refining from $(60, 801)$ to $(120, 2401)$ takes the k_y -integrated inter-plane sweep from 5.2×10^{-3} to 8.0×10^{-4} against the real-space pairwise sum (`gate_sweep_rung3_vertical`). Both refinements raise the truncation $k_{y\max}$ and the node density together, so what is demonstrated is that the quadrature converges, not which of the two limits it; the cause of the slow convergence is the same algebraic decay of the whole-space kernel costed in the accumulator argument above, and it is the reason (6) sends only the reverberation through a transverse rule.

On the arbiter that was dropped. The validation ladder originally carried one further rung: the composed operator against `FFTProp`, the faithful port of the thesis’s own Fortran, as a convergence study rather than a tolerance. That rung is *dropped*, and the reasoning is worth stating because it is about what an arbiter is for.

`FFTProp` carries no T_0 and performs no multiple-scattering solve — it propagates given scattering sources — so the “cylinder Mie T_0 against cube T_0 ” shape difference the rung was built around does not arise. Its free surface does not reproduce the free-surface reflection: two independent constructions, the Aki & Richards displacement formula and the traction-free condition applied to the stratified eigenvectors, agree with each other to six decimal places and disagree with it, its $|R_{PP}|$ staying within 1% of unity across the whole propagating range where the true surface dips to 0.15, and its P – SV conversion reaching 10^{-3} where the true one reaches 0.63. And its state vector is carried in cylindrical harmonics — which are not a coordinate choice but the shape of its *cylinders* entering the representation, where this construction discretises into cubic voxels and carries the Cartesian nine-component state throughout.

The rung’s case was provenance. Set against four arbiters that already pass — Kennett reflectivity at 10^{-15} on the marine stack, the closed-form Kupradze propagator at 3×10^{-16} , the entirely different FFT-convolution architecture at 5×10^{-16} , and a dense LU with an explicit Neumann series at 10^{-12} — a fifth reached through a lossy change of representation, against a code with a different discretisation and a free surface that does not convert, would principally measure the difference between two discretisations. That is a real quantity, and it is not validation of the sweeps.

7 Preconditioning the Krylov solve: sparsify, then sweep

The Krylov solve of (1) needs few iterations at weak contrast and very many at strong. Write $\rho = \rho(G_0 T_0)$ for the spectral radius of the scattering operator, the measure of how strongly the medium scatters. On an 8^3 lattice of cubes of pitch $p = 100$ m in a whole space ($\alpha = 5000$, $\beta = 3000 \text{ m s}^{-1}$, $\rho_0 = 2500 \text{ kg m}^{-3}$), at eight points per S wavelength ($ka = 0.24$, inside the range the analytic cube T-matrix is validated for), with a distinct contrast per voxel — $(\Delta\lambda, \Delta\mu, \Delta\rho) = (2 \text{ GPa}, 1 \text{ GPa}, 100 \text{ kg m}^{-3})$ scaled by a strength factor and by a uniform random factor in $[0.5, 1.5]$ — unpreconditioned GMRES to 10^{-8} takes 7 iterations at $\rho = 0.06$ and 998 at $\rho = 6.4$. The thesis already preconditioned this solve with a Riccati sweep (Ch. 6, Alg. 6.4); what follows asks what such a sweep must carry in three dimensions.

A caveat that governs every table in this section. The rows of each table are strength factors 1, 10, 30 and 60, in order. Against the background ($\lambda_0 = 17.5$, $\mu_0 = 22.5 \text{ GPa}$), strength 1 is $\Delta\lambda/\lambda_0 = 11\%$, $\Delta\mu/\mu_0 = 4\%$, $\Delta\rho/\rho_0 = 4\%$ — the validated test contrast. Strength 10 is already 114%, 44%, 40%, and strengths 30 and 60 reach 343–686% of λ_0 : beyond the $|\Delta| < 52\%$ validity floor of the single-site renormalisation, and beyond any physical rock. Only the first row of each table is a physical medium. At that row the preconditioner saves one or two iterations of about seven. The larger savings at $\rho \geq 0.4$ are demonstrations of the *mechanism* on a strongly scattering but unphysical operator, not of a gain on a physical problem. Strong scattering at physical contrast arises instead from the number of scatterers — ρ grows with the lattice at fixed contrast — and that regime is not yet measured here. The structural results below do not depend on the contrast: the stencils, their conditioning and the Schur-complement analysis involve G_0 alone.

7.1 The sparsified system

Liu and Ying [9] make a dense Lippmann–Schwinger system nearly local before sweeping it. For voxel i let μ_i be its $3 \times 3 \times 3$ neighbourhood (clipped at the domain faces) and μ_i^c the rest of the domain. A *stencil* α_i — a matrix of 9 columns, one per equation the voxel contributes, and $9|\mu_i|$ rows — combines the $9|\mu_i|$ rows of $(I - G_0 T_0)\psi = \psi^{\text{inc}}$ indexed by μ_i . If $\alpha_i^\dagger G_0[\mu_i, \mu_i^c] \approx 0$, the combination involves only unknowns in μ_i :

$$\alpha_i^\dagger \psi_{\mu_i} - \beta_i^\dagger T_0 \psi_{\mu_i} = \alpha_i^\dagger \psi_{\mu_i}^{\text{inc}}, \quad \beta_i^\dagger = \alpha_i^\dagger G_0[\mu_i, \mu_i], \quad (9)$$

where $\dagger$ is the conjugate transpose. Stacking (9) over all voxels gives a sparse matrix H , and stacking the combinations $\alpha_i^\dagger$ gives the stencil operator P , so that $H \approx P(I - G_0 T_0)$. The preconditioner is $M = H^{-1}P$, applied on the right. Rows are combined after scaling the strain rows by p , so that all nine have the units of displacement.

Annihilating the far field is not enough. The natural choice of α_i — the 9 left singular vectors of $G_0[\mu_i, \mu_i^c]$ with the smallest singular values — leaves far-field residuals of 1.5×10^{-3} of the largest response at eight points per wavelength, against 1.5×10^{-4} for Liu and Ying’s scalar case at the same pitch (`measure_sparsify_stencil_accuracy`). Yet the resulting preconditioner never converges: the nine combinations chosen at neighbouring voxels are not independent, P is singular (condition number 2×10^{18} on 6^3) and H nearly so, and although H matches $P(I - G_0 T_0)$ to 6×10^{-4} , H^{-1} amplifies the mismatch until the preconditioned operator has eigenvalues as small as 10^{-10} . Liu and Ying’s single scalar stencil is, in effect, a centred discrete Helmholtz operator, invertible by construction; the nine elastic stencils carry no such guarantee.

Centre-anchored stencils. Anchoring each row on its own voxel, as a finite-difference stencil is anchored on its centre, restores it. Write $\alpha_i = [I; X_i]$, with the identity on the voxel's own nine components and X_i on its neighbours, and let K_c and K_o be the rows of $G_0[\mu_i, \mu_i^c]$ belonging to the voxel and to its neighbours. The far leakage is $K_c + X_i^\dagger K_o$, minimised by

$$X_i^\dagger = -K_c K_o^\dagger \left(K_o K_o^\dagger + \lambda \|K_o K_o^\dagger\| I \right)^{-1}, \quad (10)$$

with a small Tikhonov weight λ . At $\lambda = 10^{-3}$ the leakage rises to $1\text{--}4 \times 10^{-2}$, but $P = I + (\text{neighbour blocks})$ is invertible (condition number 3×10^3) and the preconditioned spectrum stays near one. Fitting the stencils against the whole domain and solving H exactly by sparse LU, GMRES iterations on 6^3 are:

ρ	none	27-voxel truncation of G_0	centre-anchored
0.06	7	7	5
0.57	20	16	9
1.98	61	36	13
4.99	269	142	28

The truncation column — $I - G_0 T_0$ with G_0 cut to the neighbourhood, no stencils — is the control: the stencils, not the local inversion alone, carry most of the gain.

Fitted once. The stencils need not be fitted per voxel or per domain. Classify a voxel by which faces it touches — interior, face, edge or corner, 27 types — and fit one stencil per type against a fixed window of far offsets, $|\Delta|_\infty \leq 7$ pitches, clipped on the side a boundary voxel lacks: Liu and Ying's union of translated complements. The fit takes about a minute and does not depend on the domain or the contrast. It keeps nearly all of the gain, and the gain grows with the domain:

lattice	ρ	none	fitted per domain	fitted once
6^3	4.99	269	28	34
8^3	0.69	23	10	11
8^3	2.48	99	20	23
8^3	6.41	998	43	50

(measure_sparsify_preconditioner).

In a stratified background. Six depth planes one pitch apart, on half-pitch sublayers, with a material contrast half a pitch from two of them — medium A (5, 3 km s^{-1} , 2.5 g cm^{-3}) above, medium B (6.5, 3.7, 3.0) below — under water, $Q = 50$, pitch 0.25 km at 1.5 Hz, with the reverberation table at $k_r p = 20$ as the correction above requires. The operator is assembled per plane, each plane's same-depth term from its own medium. GMRES iterations:

ρ	none	27-voxel truncation	stencils, whole space (A)	stencils, layered
0.05	6	5	4	4
0.45	15	12	7	7
1.52	43	27	11	11
3.68	167	83	17	16

The stencils matter — five times fewer iterations than truncation at the strongest, unphysical, contrast — but the layering in them does not. Stencils fitted to medium A's whole-space kernel do as well as stencils fitted plane by plane to the layered kernel, although the layered operator

differs from that kernel by 43–97% in medium B. The system H carries the true local blocks through β_i either way; only the far-annihilating combinations come from the fit, and they tolerate a wrong kernel (`measure_sparsify_layered`). One whole-space stencil set therefore serves a stratified background.

The cost of solving H exactly. H is sparse, and a sparse LU of it is contrast-independent in cost. On 16^3 (36,864 unknowns, whole space) the factorisation takes 7.5 minutes with 1.8×10^8 nonzeros of fill, about 3.5 GB, and each application 0.6 s, against 3.7 s for one application of G_0 (`measure_sparse_direct_16`). A factorisation depends on the medium and the frequency, not the source, so it can be shared by every source of a survey.

7.2 What the sweep must carry

Exact LU of H does not scale. Because the stencil couples only neighbouring planes, H is block-tridiagonal in depth, with diagonal blocks D_z , sub-diagonal L_z and super-diagonal U_z , and exact LU *is* the Riccati up/down sweep of the thesis’s Alg. 5.1:

$$S_1 = D_1, \quad S_z = D_z - L_z S_{z-1}^{-1} U_{z-1}, \quad (11)$$

a downsweep forming the Schur complements S_z , then back-substitution upward. S_z is the discrete Dirichlet-to-Neumann map of the half-domain above plane z : it carries the radiation condition. On 8^3 , GMRES iterations with the sweep variants replacing exact LU (`measure_sparsify_sweep`):

ρ	none	exact	$S_z \approx D_z$	nested	reference S_z	band $r = 1/2/3$
0.06	7	6	56	63	8	35/22/13
0.69	23	11	59	66	23	37/25/17
2.48	99	23	73	87	58	50/40/32
6.41	998	50	185	279	332	229/279/302

The exact sweep reproduces exact LU at every contrast. *Dropping the Schur term*, $S_z \approx D_z$ — the literal reading of “ignore backscattering” — is worse than no preconditioner at weak contrast: without the map of the half-domain above, every plane reflects. Nesting a left/right sweep inside each plane (the full alternating-direction scheme) inherits the defect. The *reference* Schur complements, computed once from the contrast-free system, are the closer analogue of the thesis’s $\tilde{P}_F$, which keeps the reference medium’s transmission and reflection and ignores only the heterogeneity’s backscatter; they serve at weak contrast and fail at strong, where that backscatter dominates. Truncating each S_z to a lateral band of radius r pitches converges slowly in r and is not even monotone at $\rho = 6.4$: a Dirichlet-to-Neumann map is laterally nonlocal. The same object is the impedance of the continuum march, which is marched exactly for that reason.

A moving absorbing layer, and why it fails here. Liu and Ying replace S_z^{-1} by the solve of a small local problem: the last b planes down to z , closed above by a perfectly matched layer whose stencils annihilate complex-stretched plane waves. Transferred to the elastic system — P and both S waves in the nine-component state, the strain rows carrying the stretching Jacobian, fitted to 26, 64 or 78 directions — it fails (`measure_sparsify_moving_layer`, `measure_sparsify_pml`). On 6^3 every padded window preconditions worse than a plain cut, and against the exact S_z^{-1} of the contrast-free system a plain cut is already 0.78 off while the unstretched pads are 10 to 380 times off. Anchoring each pad stencil on the interior stencil of the same type, so that the unstretched pad is simply more interior medium, does not help: the discrepancy follows the *depth* of the truncated system, not the construction of the pad. The likely reason, not

yet tested, lies in how H is closed. Here the rows of voxels on the domain faces use stencils fitted to cancel the dense operator’s far field, which makes the global system equivalent to the dense one — the source of the exact-solve result above — but leaves H by itself a closed cavity: the radiation condition is carried by the cancellation between H^{-1} and P , and the Schur complements of H are cavity maps rather than Dirichlet-to-Neumann maps. Any window breaks that cancellation. Liu and Ying’s H is radiating by construction: the domain is extended on every side by layers of absorbing voxels with absorbing stencils, so its Schur complements are genuine Dirichlet-to-Neumann maps and a moving layer has something to approximate. Building H that way is the next step.

8 What is distinctive

Two choices set this apart from a standard FFT-accelerated volume integral equation. First, the forward operator is a directional partial-wave summation rather than a circular-convolution FFT, so the in-plane coupling is summed along the separated axes and the $\Delta z = 0$ divergence is structurally absent rather than regularised — and where the stratified background does contribute at equal depth, §2.4, it enters as a difference in which the divergence cancels rather than as a term requiring regularisation. *In three dimensions the first of these changes character:* by §6 the same-depth coupling is a direct real-space sum of the closed form rather than a directional summation, and what survives of the partial-wave structure is the depth direction and the reverberation. The absence of any periodicity assumption survives intact, and by a stronger route — a real-space table assumes none at all, where a transverse quadrature merely avoids introducing one. Second, the depth structure is carried exactly by the stratified propagator on the true layered background, not approximated on a homogeneous whole-space kernel. The multiple scattering is left where it belongs, in a Krylov solve that calls the sweep as its matvec, so the propagator stays a clean, stable, forward summation and the solver carries the contrast.

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